



VietNam National University University of Engineering and Technology

EMBEDDED SYSTEM FUNDAMENTALS (ELT3240, NHẬP MÔN HỆ THỐNG NHÚNG)

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Laboratory for Smart Integrated Systems



**VietNam National University
University of Engineering and Technology**

COURSE OVERVIEW

Laboratory for Smart Integrated Systems

General Information

- **Instructor**
 - Hung Nguyen-Kiem, PhD
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 - Email: kiemhung@vnu.edu.vn

Objectives

- Put a large emphasis on the architecture of ARM-based embedded computing systems
 - Emphasis on ARM Cortex-M embedded processor micro-architectures (via the instruction set architecture) and ARM-based SoCs
 - Emphasis on the process and techniques for design hardware as well as programming and interfacing embedded systems with real-world devices, embedded EDA tool chains, and application design
 - Demonstrating through practical examples
- Practices and a course project used to help you put the ideas from the course into practice

Objectives

After completing this course, you will master:

- Essential knowledge of embedded systems,
- The architecture of ARM processors and ARM-based embedded SoCs,
- Techniques for developing both hardware and software components for ARM-based embedded SoCs,
- Using a typical toolchain (such as Keil, Mbed) to implement and test ARM-based embedded applications in C and assembly language
- Using a typical toolchain (such as Proteus, Altium Designer) to design and implement hardware

Contents

- **Topic 1: Introduction to Embedded Computing System**
 - What is embedded computing system?
 - Applications of embedded computing system,
 - ARM Development Environment,
 - C language for Embedded System

Contents

- **Topic 2: Hardware architecture**
 - A brief of computer architecture taxonomy
 - Architecture of an embedded system: CPU, buses, I/O devices, memory and interfaces, ...
 - ARM Cortex-M processor family and assembly language
 - ARM-Based Embedded System-on-Chips (SoCs)

Contents

- **Topic 3: Embedded Software**
 - The processes for compiling, assembling, and linking software components,
 - Developing some embedded software components
 - Basic concepts of processes and process-based design: Schedule and Inter-process communication,
 - Real-time operating systems (RTOS)

Contents

- **Topic 4: Interfacing to ARM Cortex-M microcontrollers**
 - I/O Programming based on GPIO ports
 - Interfacing and programming of serial communication standards such as UART, I2C, SPI,...
 - ADC/DAC and sensor/actuator interfacing
 - The real-time programming with the Interrupt mechanism of the ARM Cortex-M microcontrollers
 - ...

Contents

- **Topic 5: Embedded Platform Design**
 - Embedded computer platform design method
 - Hardware design support tools: Proteus, Altium Designer
 - Schematic design
 - PCB circuit design and manufacturing

Contents

- **Topic 6: Practices and Projects**
 - **Labs:** Write programs for a range of typical microcontroller peripherals (e.g. GPIO, USART, I2C, ADC, Sensors, etc...) of **STM32 Nucleo-F401re board**
 - **Assignments 1&2:** designing and implementing simple embedded systems, including both hardware and software, based on ARM Cortex-M microcontroller using tools Keil and Proteus
 - **Final Project:** Design and build specialized embedded systems for a specific purpose

Prerequisites

- Basic computer programming knowledge (e.g. C , Assembly language)
- Digital design knowledge and experience
- Basic knowledge of processor

Resources

- ARM
 - EDA/CAD tools: **Keil MDK-ARM IDE (v5)**
 - <https://www.keil.com/download/product/>
 - Evaluation Kit: **STM32 Nucleo F401re board**
 - Others: **Documents, AppNote, datasheet, Examples,...**
 - **My blog:** <https://dayhocstem.com/blog/dao-tao/he-thong-nhung>
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Teaching materials

- **Textbook(s):**

- Lecture Slides
- Alexander G Dean: “Embedded Systems Fundamentals with ARM Cortex-M based Microcontrollers: A Practical Approach”, ARM Education Media (2017).
- Ünsalan, C., Gürhan, H. D., & Yücel, M. E. (2022). Embedded System Design with ARM Cortex-M Microcontrollers. Springer International Publishing.

- **Reference books/materials:**

- Trevor Martin: “The Designer’s Guide to the Cortex-M Processor Family”, Elsevier Ltd (2013).
- Joseph Yiu: “The Definitive Guide to the ARM Cortex-M0”, Elsevier Ltd (2011)

Administrative Details

- Course Grading
 - Class Quiz and Participation: 10%
 - Regular Test: 10%
 - Mid-term exam (Multi-Choice Quiz): 20%
 - Final Exam (Oral on Practices and Project): 60%
- Students have to be present:
 - at least 80% of the course meetings

Lectures, Homework, Quizzes

- It is important to attend each lecture
 - Lots of the material covered in lecture is not in the book or readings
- There will be several homework assignments
 - Not all will be collected/graded
 - May require use of C/C++ and Assembly
- There will be in-class quizzes

Project [1]

- For the course project, you will be expected to complete original work related to embedded systems in teams of **two** or **three** students.
- Projects will consist of **source code**, **final report**, and short presentation.
- You are encouraged to come up with your own topic for the project, but will be provided with a list of possible ideas.

Project [2]

- The course project counts for 60% of the overall grade.
- Students will be expected to turn in a summary of the contribution to the project to facilitate fair grading to individual team members.
- Additional details on the course projects will be provided in a separate document.

Honor

- You are encouraged to collaborate with other students in projects
- Final practice and project report, homework should be done by yourself
- Final Exam is closed book, closed notes... (only pen, blank paper are allowed)



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Thank You

Questions?

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